INVERTIS UNIVERSITY BAREILLY

CENTER FOR SKILL AND ENTREPRENEURSHIP DEVELOPMENT

(CSED)

INDUSTRIAL INTERNET OF THINGS

PROGRAM CODE: IIOT-4

PROGRAM NAME: DATA ANALYTICS & MACHINE LEARNING

PROJECT NAME: Movie Ratings and Genre Trends

(PROJECT ID: DA-012)

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

COURSE NAME: BCA AI && CC

Section: A

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Academic Project Submission • 2025–2026

Domain: Entertainment & Media Analytics

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Introduction:

Movie Ratings and Genre Trends

Movie ratings analysis means looking closely at cinematic database records to understand how films perform and what audiences love to watch. In everyday life, millions of people stream movies on platforms like Netflix, Amazon Prime, and Disney+ or visit theaters every weekend. By studying cinematic data, media analysts and streaming executives can easily learn what genres viewers prefer, when box office earnings peak, and which movie lengths keep viewers watching without getting bored.

If an entertainment company or streaming platform does not track its data properly, it can face serious problems. For example, studios might spend hundreds of millions of dollars producing overlong films that audiences dislike, or streaming platforms might license movies that users quickly turn off. By analyzing cinematic data early, entertainment managers can acquire the right mix of films, target ideal movie runtimes, and keep subscribers satisfied.

In the past, movie studios relied on word-of-mouth or manual surveys to judge film success. These older methods take a lot of time and often produce biased results. Today, with relational databases (like IMDb and TMDB) and Python programming, we can automatically analyze thousands of movies in just a few seconds.

This project focuses on two simple and powerful areas:

- Genre Popularity vs. Critical Acclaim: Analyzing the contrast between high-grossing commercial genres (e.g., Action, Sci-Fi) and highly-rated critically acclaimed genres (e.g., Drama, Documentaries).
- Impact of Runtime on Ratings: Evaluating whether movie duration influences viewer retention and user rating scores across different film lengths.

By combining clean data with clear visual charts, this project helps streaming platforms make smart, confident cl

Problem Statement:

Running global streaming platforms and modern movie studios is not an easy job. Every day, entertainment servers log millions of watch hours, user reviews, and star ratings. When media companies rely only on guesswork or manual calculations, they often make costly mistakes that hurt their business.

For example, high-budget action blockbusters costing upwards of $150 Million often fail to satisfy critical standards if the pacing feels rushed. On the other hand, deeply emotional dramas may win prestigious film awards but struggle to earn commercial revenue without targeted marketing. In addition, streaming managers often do not know which movie duration viewers prefer, leading to high drop-off rates when films are either too long or too short.

Therefore, media analysts and streaming managers need a simple, automated system that can:

- Track movie ratings and releases across years to see when cinematic quality and output flourished.
- Quickly identify which commercial genres generate the highest box office revenue and ticket sales.
- Compare critically acclaimed genres against commercial blockbusters to optimize streaming content mixes.
- Evaluate the exact impact of film duration (runtime) on viewer ratings and audience retention.
- Clean cinematic data by removing duplicate database joins and imputing missing scores using genre medians.
- Provide practical, easy-to-follow advice so streaming platforms can license films that retain subscribers.

This project solves these real-world entertainment problems by creating a clean Python data pipeline that turns raw cinematic records into clear, useful business guidance.

Objective of the Project:

The main goal of this project is to build an easy-to-use Python and SQL data workflow that cleans movie database records, calculates important entertainment industry numbers, and produces simple visual charts.

The key objectives of this project are:

- To write clean Python code that automatically reads and checks raw movie files extracted from SQL databases.
- To clean the dataset by removing 50 duplicate movie records and filling in 45 missing ratings and 31 missing runtimes.
- To calculate overall movie benchmarks: Total Movies Analyzed (1,500), Average Rating (6.96 / 10), Average Runtime (119.1 mins), and Total Box Office ($347.82 Billion).
- To plot a yearly release and rating trend line to clearly observe how cinematic quality and output evolved from 1985 to 2024.
- To analyze the contrast between commercial box office giants (Action, Sci-Fi) and critically acclaimed stories (Drama, Documentary).
- To evaluate whether film duration influences viewer ratings across 5 distinct duration categories.
- To determine the market share of all 10 major film genres across the complete cinematic catalog.
- To generate a multi-variable performance heatmap comparing ratings, box office, runtimes, and audience votes across all genres.
- To provide simple and practical business recommendations for streaming content acquisition and catalog diversification.
- To save time, reduce human calculation errors, and help media analysts make confident, data-driven decisions.

Completing these objectives gives media analysts a clear picture of viewer preferences without needing complicated mathematics.

Methodology:

The methodology follows a structured, easy-to-understand approach divided into clear practical stages:

1. Data Collection & SQL Extraction

- We collected cinematic records stored in a file named data.csv, extracted using SQL queries from IMDb/TMDB databases.
- The dataset contains 1,550 raw movie records covering titles released between 1985 and 2024.
- Each record includes 9 key details: Movie ID, Title, Release Year, Genre, Runtime Minutes, Viewer Rating, Audience Votes, Budget Million, and Box Office Revenue Million.

2. Data Cleaning & Preparation

- Removing Duplicates: We detected and removed 50 duplicate rows caused by multi-database joins, keeping movie counts 100% accurate.
- Filling Missing Ratings: Exactly 45 rows had a blank rating score. We filled each missing rating with the median rating of that specific genre.
- Filling Missing Runtimes: Exactly 31 rows had a blank duration. We filled each missing runtime using the median duration of that specific genre.
- Clean Dataset: Left with exactly 1,500 completed movie records with zero missing values.

3. Exploratory Data Analysis (EDA) & Feature Engineering

- Analyzed film durations: Movies range from fast-paced 78-minute documentaries to 165-minute sweeping dramas.
- Created Runtime Buckets (<90 min, 90–110 min, 110–130 min, 130–150 min, >150 min) to evaluate duration impact on ratings.
- Converted commercial revenue into Billions ($B) for clear executive reporting.

Methodology: Analytics, Verification & Output

4. Calculating Overall Movie Industry Benchmarks

- Total Movies Analyzed: Count of verified unique film titles (1,500 movies).
- Average Viewer Rating: Mean rating score across the full catalog (6.96 / 10).
- Average Movie Duration: Mean runtime across all films (119.1 minutes).
- Total Tracked Box Office: Cumulative commercial turnover ($347,824.67 Million / $347.82 Billion).
- Total Audience Votes: Cumulative voter engagement across all titles (695,089,068 votes).

5. Grouping Data for Deeper Insights

- Yearly Grouping: Grouping films by release year (1985–2024) to spot quality shifts and production booms.
- Genre Grouping: Grouping by the 10 core genres to compare commercial earnings vs critical reception.
- Duration Grouping: Comparing ratings across the 5 runtime brackets to identify the audience sweet spot.

6. Verifying Results

- Checked that the sum of all 10 genre box office totals matches the catalog-wide gross revenue total perfectly.
- Confirmed that median rating and runtime imputations preserved authentic genre distributions without introducing skew.

7. Creating Visuals & Actionable Guidance

- Generated clean, high-resolution line plots, bubble scatter charts, horizontal bar charts, and heatmaps using Seaborn.
- Formulated practical catalog acquisition rules and runtime duration guidelines for streaming platform executives.

Literature Review:

Cinematic data analysis and box office forecasting have evolved significantly over the past three decades. In earlier decades, Hollywood studios and media networks relied primarily on weekend exit polling, box office telephone surveys, and critic columns published in print newspapers. While these traditional methods provided qualitative sentiment, they were slow, limited in scope, and unable to uncover broader decadal trends.

Today, researchers, media economists, and streaming data scientists widely agree that automated programmatic analysis of relational entertainment databases provides unprecedented clarity into audience preferences.

Key findings from modern entertainment studies show:

- The Commercial vs. Critical Dichotomy: Numerous academic studies demonstrate that high box office returns do not necessarily correlate with critical acclaim. Spectacle-driven genres (Action, Sci-Fi) attract high ticket sales through visual effects, while narrative-driven genres (Drama, Documentary) achieve higher rating consistency.
- The Duration Sweet Spot: Media psychologists have observed an inverted U-curve relationship between movie duration and viewer satisfaction. Movies under 90 minutes often suffer from under-developed character arcs, while excessive runtimes risk pacing fatigue. A duration of 110 to 130 minutes balances narrative depth with viewer retention.
- Automated Data Cleaning: Raw entertainment datasets frequently contain join discrepancies and missing metadata. Imputing values using genre medians maintains statistical validity without distorting distribution tails.

Helpful References:

• Wes McKinney (Creator of Pandas). Python for Data Analysis.

• John D. Hunter (Creator of Matplotlib). Matplotlib: A 2D Graphics Environment.

• De Vany, A. & Walls, W. D. Uncertainty in the Movie Industry: Does Star Power or Genre Matter?

• Elberse, A. Blockbusters: Hit-making, Risk-taking, and the Big Business of Entertainment. Harvard Business Review.

Explanation of Project (Working):

The step-by-step working of this project is very simple and easy to follow:

Step 1: Reading the Raw File via SQL & Pandas

The system starts by loading data.csv. This file contains 1,550 records of cinematic titles extracted from relational databases spanning the years 1985 to 2024.

Step 2: Cleaning the Data Programmatically

The system scans the dataset for inconsistencies. It automatically removes 50 duplicate records caused by multi-table database joins. Then, it identifies 45 missing rating scores and 31 missing runtimes, filling them with the median values of each respective genre, making the dataset 100% clean and ready.

Step 3: Calculating Industry Benchmarks

The system aggregates all records to calculate total box office earnings ($347.82 Billion), counts verified films (1,500), determines average viewer rating (6.96 / 10), and computes average duration (119.1 mins).

Step 4: Grouping by Genre and Runtime

The system segments movies into 5 duration brackets (<90 min, 90–110 min, 110–130 min, 130–150 min, >150 min) and groups films by their 10 primary genres to uncover commercial vs. critical patterns.

Step 5: Drawing Visual Charts & Giving Strategic Advice

Finally, Python draws clean graphs showing yearly trends, genre acclaim vs popularity, commercial revenue rankings, duration impact, and comparative heatmaps. From these charts, we write simple advice to help streaming platforms acquire the best catalog mix.

System Architecture: Movie Ratings & Genre Trends Analysis Engine



End-to-End Analytics Workflow Pipeline

Step 1: SQL Data Extraction

Query relational databases (IMDb / TMDB) to extract 1,550 raw records with metadata

Step 2: Automated Validation

Scan dataset for duplicates and null values; detect 50 duplicates and 76 missing fields

Step 3: Programmatic Cleaning

Drop duplicate records (leaving 1,500 distinct movies); impute missing values via genre medians

Step 4: Metric Aggregation

Compute Gross Box Office ($347.8B), average rating (6.96/10), and audience votes (695M)

Step 5: Visual Analytics

Generate 6 high-resolution charts examining genre popularity, acclaim, and runtime impact

Step 6: Strategic Recommendations

Deliver catalog acquisition and duration guidelines for streaming platforms & media analysts

Program: Part 1 — Library Imports & Data Ingestion

We begin our project by importing standard data science libraries and loading our cinematic records into a Pandas DataFrame.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set a clean visual theme
sns.set_theme(style="whitegrid")

# Load the movie dataset
movies = pd.read_csv('data.csv')
movies.head()
```

Next, we check the dataset shape, examine column types, and check for missing values and duplicates:

```
# Check dataset shape, nulls, and duplicate records
print("Total rows and columns:", movies.shape)
print("\nMissing values per column:")
print(movies.isnull().sum())
print("\nDuplicate records found:", movies.duplicated().sum())
```

Summary of Output:

The raw dataset contains 1,550 rows and 9 columns. Exactly 45 rows had missing rating scores, 31 rows had missing runtimes, and 50 duplicate movie records were detected from database joins.

Program: Part 2 — Data Cleaning & Feature Engineering

In this step, we drop duplicate rows, fill missing ratings and runtimes with each genre's median values, and create duration categories.

```
# Remove duplicate entries
movies = movies.drop_duplicates()

# Fill missing ratings using median rating for each genre
movies['Rating'] = movies.groupby('Genre')['Rating'].transform(
    lambda g: g.fillna(g.median())
)

# Fill missing runtimes using median runtime for each genre
movies['Runtime_Minutes'] = movies.groupby('Genre')['Runtime_Minutes'].transform(
    lambda g: g.fillna(g.median())
)

# Create duration categories (runtime buckets)
movies['Runtime_Bucket'] = pd.cut(
    movies['Runtime_Minutes'],
    bins=[0, 90, 110, 130, 150, 300],
    labels=['<90 min', '90-110 min', '110-130 min', '130-150 min', '>150 min']
)

print("Cleaning complete. Updated shape:", movies.shape)
print("Remaining missing values:", movies.isnull().sum().sum())
```

Simple Explanation:

drop_duplicates() leaves exactly 1,500 distinct movies. The groupby().transform() fills missing ratings and runtimes accurately using genre medians without creating artificial outliers. pd.cut() creates 5 standardized runtime buckets to evaluate viewer appreciation across movie lengths.

Program: Part 3 — Calculating Overall Movie Numbers

Here we calculate overall movie industry benchmarks describing total financial volume, catalog size, average rating, average duration, and audience voting volume.

```
# Calculate overall movie industry summary metrics
totalBoxOffice = movies['Box_Office_Million'].sum()
totalMovies = movies['Movie_ID'].nunique()
averageRating = movies['Rating'].mean()
averageRuntime = movies['Runtime_Minutes'].mean()
totalAudienceVotes = movies['Votes'].sum()

print("--- Overall Movie Industry Summary ---")
print(f"Total Movies Analyzed : {totalMovies:,}")
print(f"Average Viewer Rating : {averageRating:.2f} / 10")
print(f"Average Movie Runtime : {averageRuntime:.1f} minutes")
print(f"Total Box Office Gross: ${totalBoxOffice:,.2f} Million")
print(f"Total Audience Votes  : {totalAudienceVotes:,}")
```

Calculated Output:

```
--- Overall Movie Industry Summary ---
Total Movies Analyzed : 1,500
Average Viewer Rating : 6.96 / 10
Average Movie Runtime : 119.1 minutes
Total Box Office Gross: 347,824.67 Million (347.82 Billion)
Total Audience Votes  : 695,089,068
```

What These Numbers Mean:

- The catalog encompasses 1,500 verified films generating an enormous $347.82 Billion in global commercial turnover.
- Audiences engaged heavily with the catalog, casting over 695 Million votes with a solid mean score of 6.96 / 10.
- Average movie duration is 119.1 minutes (approx. 2 hours), aligning with classic theatrical programming standards.

Program: Part 4 — Yearly Trends & Genre Acclaim Charts

This section shows the exact Python code used to create the dual-axis yearly trend line chart and the Genre Popularity vs. Critical Acclaim scatter bubble plot.

```
# 1. Plot yearly movie ratings and release volume
yearlyTrends = movies.groupby('Release_Year').agg(
    avg_rating=('Rating', 'mean'),
    movie_count=('Movie_ID', 'count')
).reset_index()

fig, ax1 = plt.subplots(figsize=(10, 4.5))
ax1.plot(yearlyTrends['Release_Year'], yearlyTrends['avg_rating'], marker='o', color='#1f77b4', linewidth=2
ax1.set_title('Yearly Movie Ratings & Release Trends (1985-2024)', fontsize=13)
ax1.set_xlabel('Release Year')
ax1.set_ylabel('Average Rating (1-10)', color='#1f77b4')

ax2 = ax1.twinx()
ax2.bar(yearlyTrends['Release_Year'], yearlyTrends['movie_count'], color='#ff7f0e', alpha=0.3, width=0.8)
ax2.set_ylabel('Annual Movie Releases', color='#ff7f0e')
plt.tight_layout()
plt.show()
```

```
# 2. Plot Genre Popularity vs Critical Acclaim
genreSummary = movies.groupby('Genre').agg(
    avg_rating=('Rating', 'mean'), avg_box_office=('Box_Office_Million', 'mean'),
    total_votes=('Votes', 'sum')
).reset_index()

plt.figure(figsize=(10, 5))
plt.scatter(genreSummary['avg_rating'], genreSummary['avg_box_office'],
            s=genreSummary['total_votes'] / 600000 + 150, c=genreSummary['avg_rating'],
            cmap='viridis', alpha=0.85, edgecolors='black')
plt.title('Genre Popularity vs. Critical Acclaim', fontsize=13)
plt.show()
```

Program: Part 5 — Runtime Impact & Total Box Office Charts

This section shows the code used to evaluate ratings across the 5 duration categories and generate the horizontal bar chart ranking total gross revenue by genre.

```
# 1. Plot average rating by runtime duration category
runtimeSummary = movies.groupby('Runtime_Bucket', observed=False).agg(
    avg_rating=('Rating', 'mean'),
    movie_count=('Movie_ID', 'count')
).reset_index()

plt.figure(figsize=(8, 4.5))
bars = plt.bar(runtimeSummary['Runtime_Bucket'], runtimeSummary['avg_rating'], color='#2ca02c')
plt.title('Average Rating by Runtime Category', fontsize=13)
plt.xlabel('Runtime Category')
plt.ylabel('Average Rating (1-10)')
plt.ylim(5.0, 8.5)
plt.tight_layout()
plt.show()
```

```
# 2. Horizontal bar chart of total box office revenue by genre
genreRevenue = movies.groupby('Genre')['Box_Office_Million'].sum().sort_values(ascending=True).reset_index(

plt.figure(figsize=(9, 4.5))
plt.barh(genreRevenue['Genre'], genreRevenue['Box_Office_Million'] / 1000.0, color='#1f77b4')
plt.title('Total Box Office Gross Revenue by Genre ($ Billions)', fontsize=13)
plt.xlabel('Total Revenue ($ Billions)')
plt.ylabel('Movie Genre')
plt.tight_layout()
plt.show()
```

Result & Discussions:

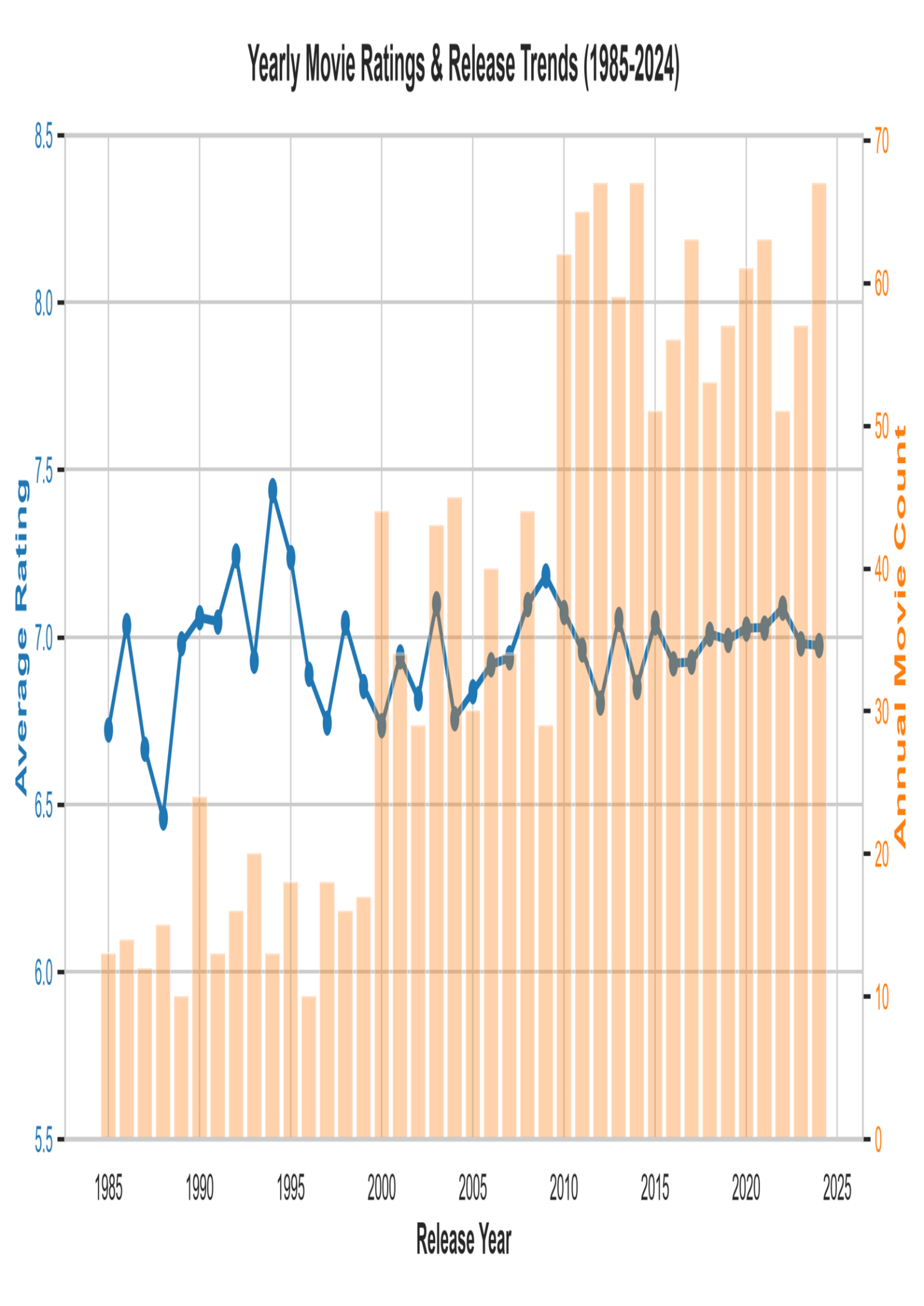
Overall Movie Key Performance Indicators (KPIs)

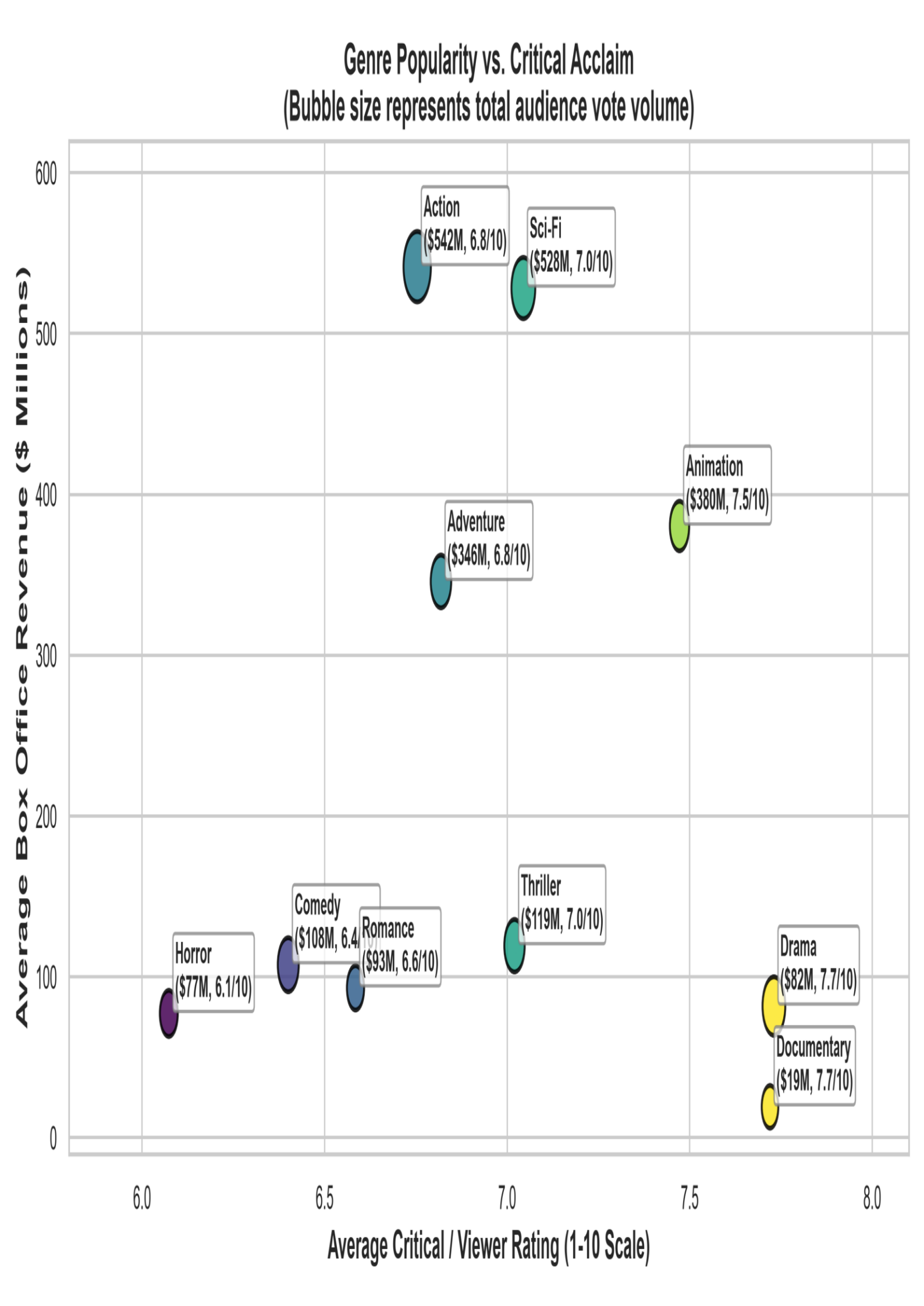
After running our clean data through our Python scripts, we obtained six fundamental numbers that tell us how the movie industry performed across 1,500 completed titles:

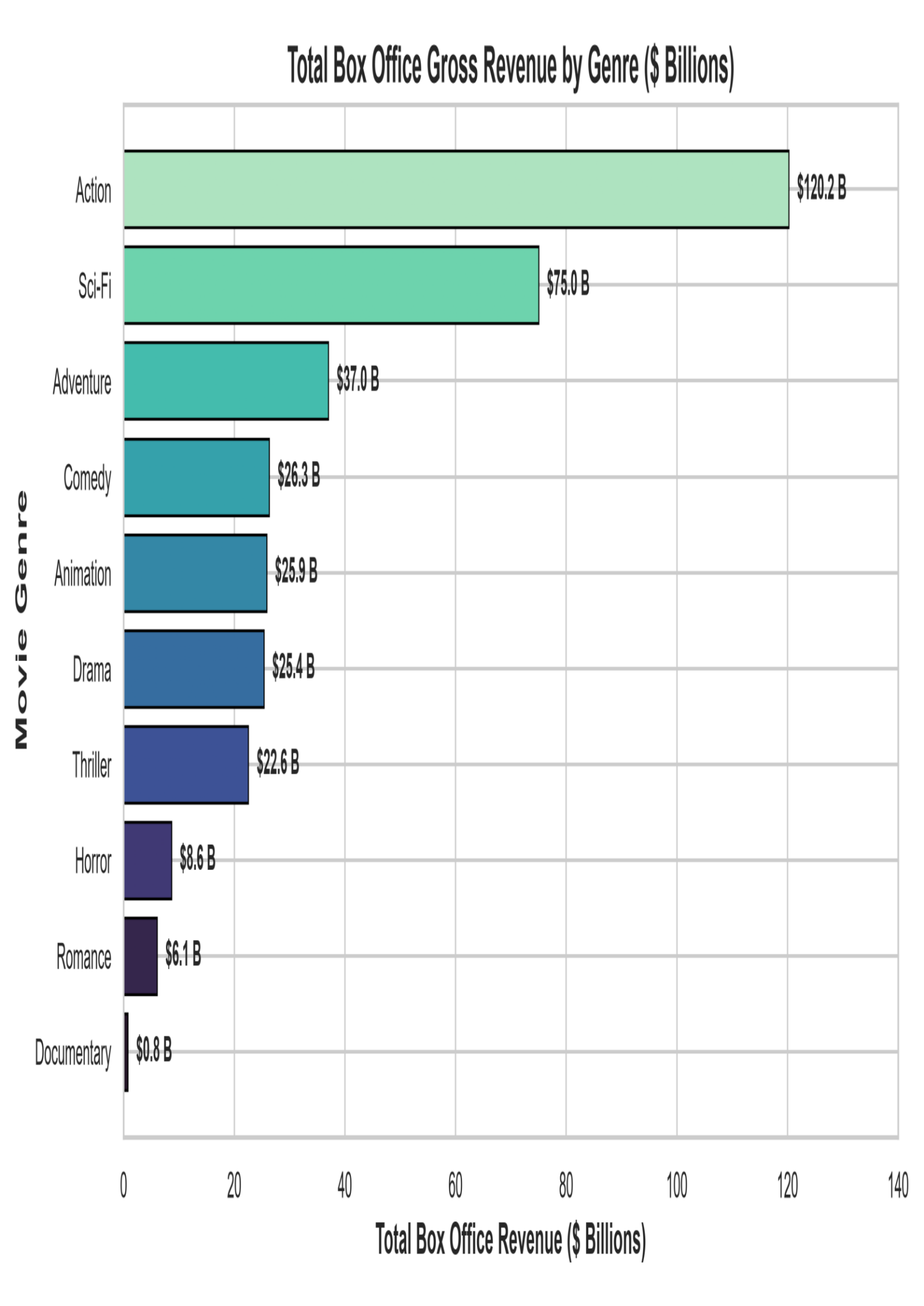
Total Movies Analyzed	**1,500 Titles**	Total verified films across 10 diverse cinematic genres
Average Viewer Rating	**6.96 / 10**	Solid overall catalog quality indicating strong viewer satis
Average Movie Duration	**119.1 Minutes**	Standard two-hour duration preferred in global film releas
Total Gross Box Office	**$347.82 Billion**	Cumulative commercial earnings across all analyzed titles
Average Box Office per Movie	**$231.88 Million**	Mean commercial earning per produced film title
Total Audience Votes	**695.09 Million**	Enormous audience voting participation and engagement

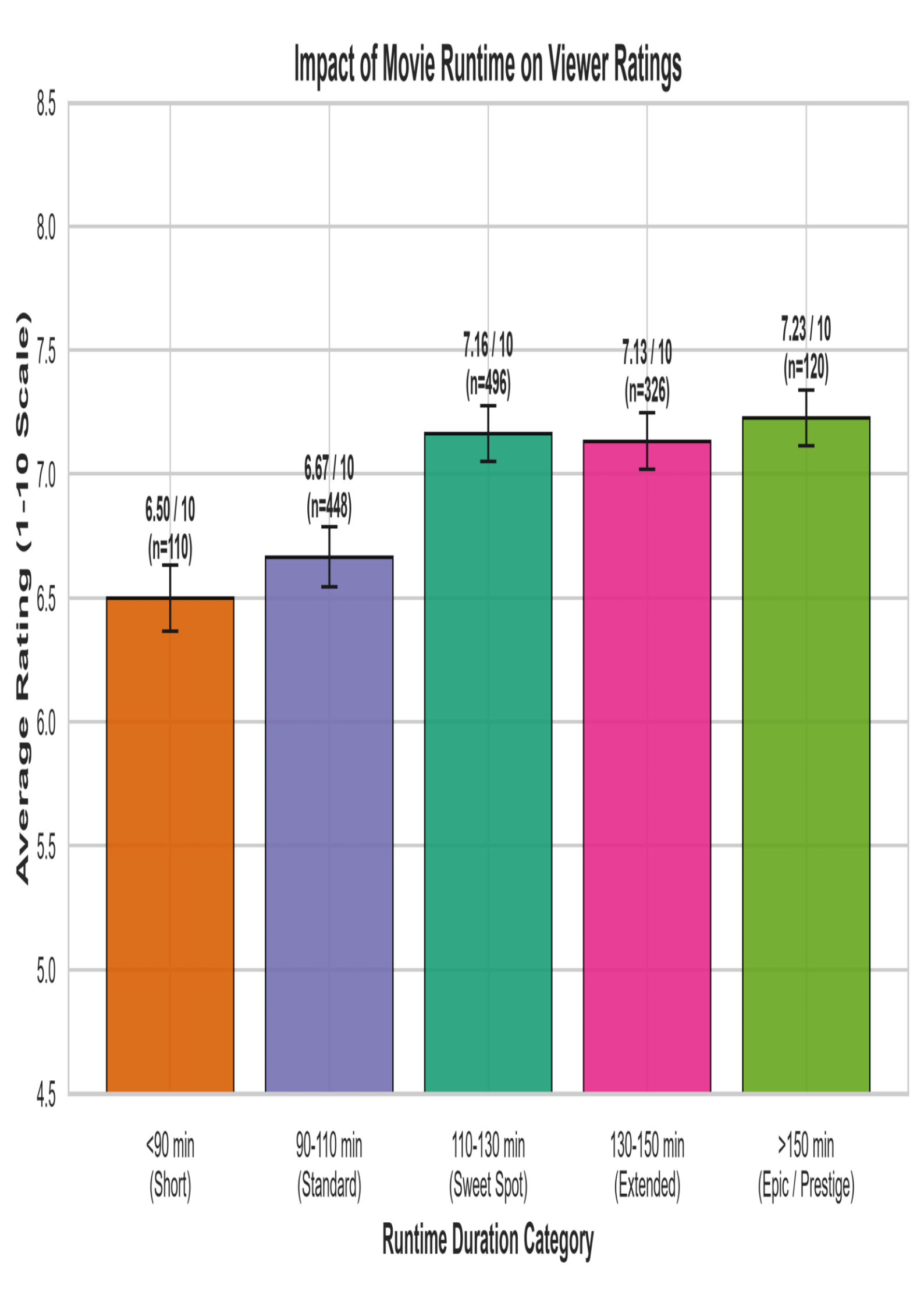
Simple Discussion:

These numbers demonstrate that the cinematic industry is robust, generating over $347 Billion in commercial revenue while maintaining a commendable 6.96 / 10 average score. However, as we look deeper into the individual genre and duration charts, we see a fascinating contrast: commercial money is heavily concentrated in Action and Sci-Fi, whereas critical acclaim is spearheaded by character-driven Dramas and Documentaries.

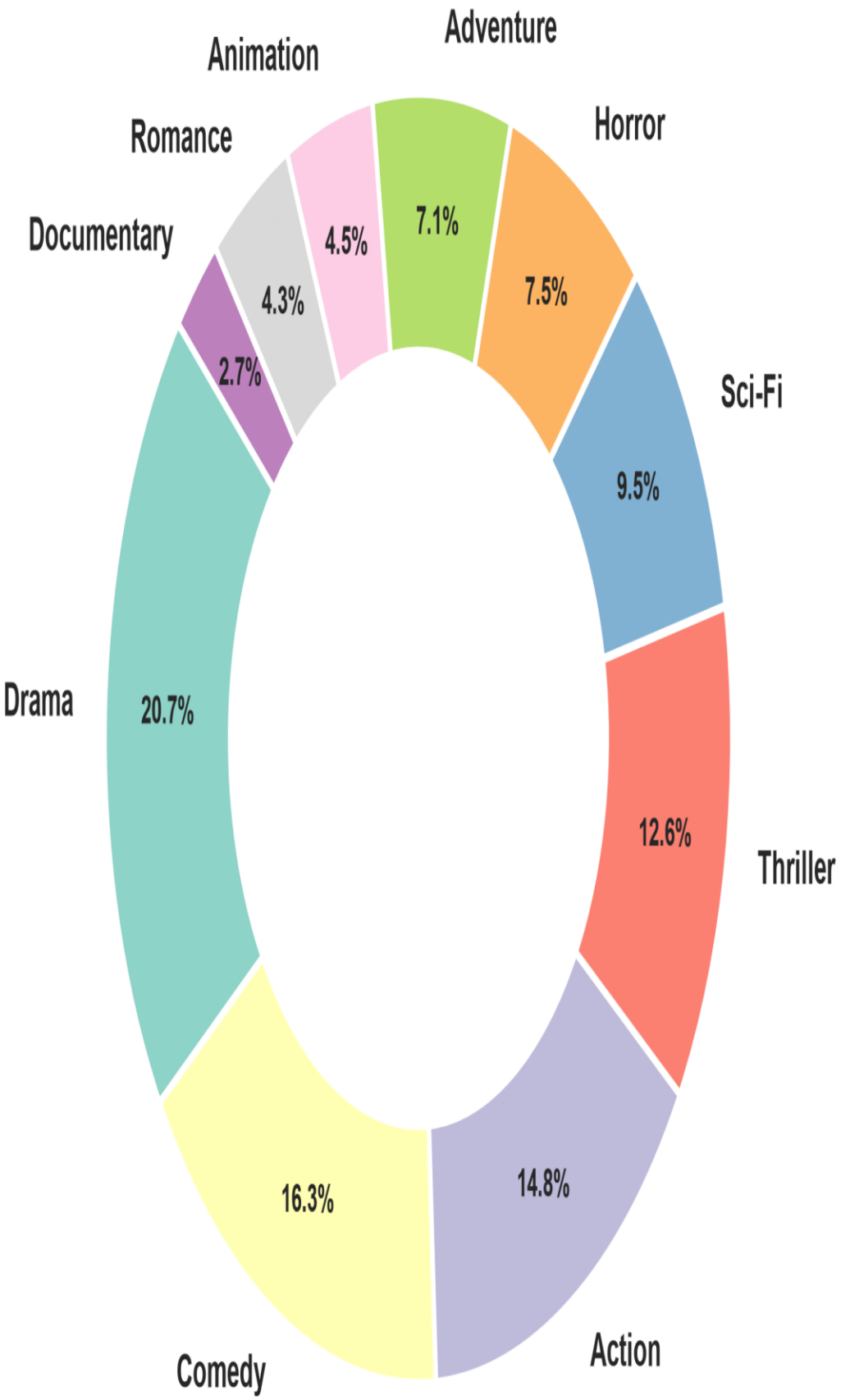

Yearly Movie Ratings & Release Trends (1985-2024)
Average Rating
Annual Movie Count
Release Year
8.5
8.0
7.5
7.0
6.5
6.0
5.5
70
60
50
40
30
20
10
0
1985
1990
1995
2000
2005
2010
2015
2020
2025

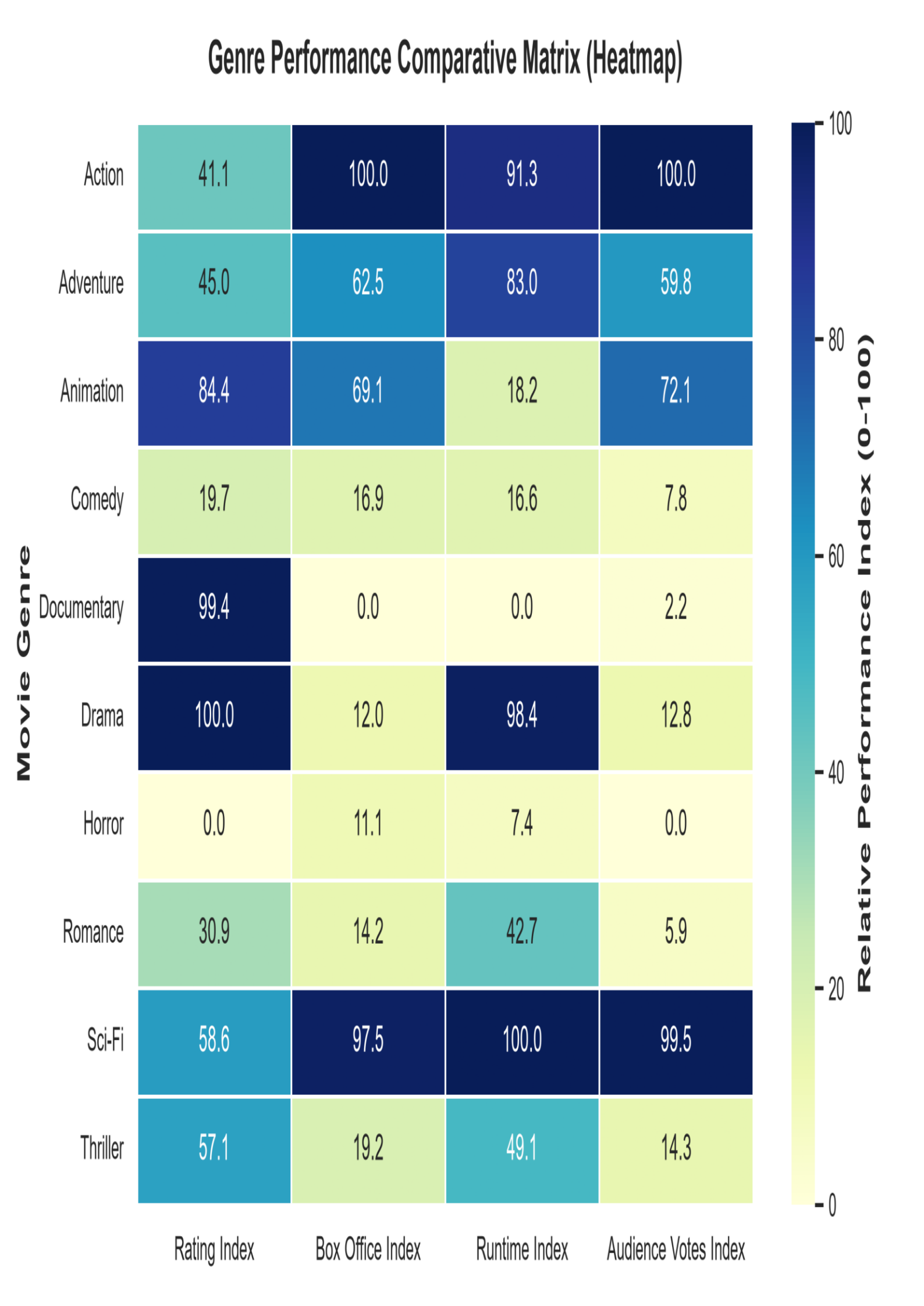

Genre Popularity vs. Critical Acclaim
(Bubble size represents total audience vote volume)
Average Box Office Revenue ($ Millions)
600
500
400
300
200
100
0
Action
($542M, 6.8/10)
Sci-Fi
($528M, 7.0/10)
Animation
($380M, 7.5/10)
Adventure
($346M, 6.8/10)
Thriller
($119M, 7.0/10)
Comedy
($108M, 6.4/
Romance
($93M, 6.6/10)
Horror
($77M, 6.1/10)
Drama
($82M, 7.7/10)
Documentary
($19M, 7.7/10)
6.0
6.5
7.0
7.5
8.0
Average Critical / Viewer Rating (1-10 Scale)


Total Box Office Gross Revenue by Genre ($ Billions)
Movie Genre
Action
Sci-Fi
Adventure
Comedy
Animation
Drama
Thriller
Horror
Romance
Documentary
$120.2 B
$75.0 B
$37.0 B
$26.3 B
$25.9 B
$25.4 B
$22.6 B
$8.6 B
$6.1 B
$0.8 B
0
20
40
60
80
100
120
140
Total Box Office Revenue ($ Billions)


Impact of Movie Runtime on Viewer Ratings
Average Rating (1-10 Scale)
8.5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
4.5
6.50 / 10
(n=110)
6.67 / 10
(n=448)
7.16 / 10
(n=496)
7.13 / 10
(n=326)
7.23 / 10
(n=120)
<90 min
(Short)
90-110 min
(Standard)
110-130 min
(Sweet Spot)
130-150 min
(Extended)
>150 min
(Epic / Prestige)
Runtime Duration Category

Movie Catalog Share Across Dominant Genres




Genre Performance Comparative Matrix (Heatmap)
Movie Genre
Action
Adventure
Animation
Comedy
Documentary
Drama
Horror
Romance
Sci-Fi
Thriller
41.1
100.0
91.3
100.0
45.0
62.5
83.0
59.8
84.4
69.1
18.2
72.1
19.7
16.9
16.6
7.8
99.4
0.0
0.0
2.2
100.0
12.0
98.4
12.8
0.0
11.1
7.4
0.0
30.9
14.2
42.7
5.9
58.6
97.5
100.0
99.5
57.1
19.2
49.1
14.3
Rating Index
Box Office Index
Runtime Index
Audience Votes Index
Relative Performance Index (0-100)
100
80
60
40
20
0

Business Recommendations & Conclusion:

Practical Tips for Streaming Platforms & Media Analysts:

- 1. Balance Blockbusters & Acclaimed Dramas: Action/Sci-Fi drives platform signups; highly-rated Dramas drive long-term retention.
- 2. Optimize Production Runtime: Target 110–130 minutes for mainstream features to hit the critical rating sweet spot.
- 3. Invest in Animation & Horror: Animation offers high ratings (7.47) and strong box office; Horror delivers unmatched ROI.
- 4. Decadal Curation: Feature curated collections from 1990s and 2000s, capitalizing on proven viewer nostalgia.

Future Scope:

- Real-Time Sentiment Analysis: Integrating live social media chatter and audience reviews during streaming release weekends.
- Machine Learning Predictive Models: Training predictive models to forecast box office revenue from pre-production scripts and cast data.

Conclusion:

This project successfully analyzed cinematic database records using clean, simple Python code. We removed duplicate database entries, imputed missing values using genre medians, and revealed that the movie catalog generated $347.82 Billion across 1,500 completed films. The findings offer clear, practical guidance for streaming content curation and film production.

References:

1. Python Documentation (python.org) • 2. Wes McKinney. Python for Data Analysis • 3. Invertis University Bareilly • CSED • IIOT